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ELECTROMAGNETIC FORM FACTORS OF THE PROTON AT LOW FOUR-MOMENTUM TRANSFER (II)

F. BORKOWSKI, G.G. SIMON, V.H. WALTHER and R.D. WENDLING*
Institut für Kernphysik der Universität Mainz, D-65 Mainz, Germany

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The 300 MeV electron linear accelerator at Mainz has been used to measure the angular dependence of the electron-proton elastic scattering cross sections at seven different energies for squared four-momentum transfers between 0.13 and 4.7 fm^{-2}. The proton form factors have been extracted from the cross sections by means of Rosenbluth plots and by fitting parametrized analytical functions directly to the cross sections. The best fit is compared to the data of other laboratories. The previously reported deviations from the dipole fit have been confirmed. From the form factors at $q^2 < 0.9$ fm^2 the proton r.m.s. radius has been determined. A determination of the spectral function of the nucleon isovector form factor G_E^V in the time-like region is obtained using a realistic ρ resonance.

1. Introduction

The understanding of the hadronic structure of particles and nuclei is a fundamental problem of strong interaction physics. The hadron to study most easily is the proton and the method which is frequently used to study the proton is the method of elastic electron scattering. The cross sections of the elastic electron-proton scattering for single photon exchange are given by the Rosenbluth formula [1] which relates a single cross section to two form factors, for instance the electric and the magnetic form factor G_E and G_M introduced by Sachs [2]. The form factors are pure functions of q^2, the squared four-momentum carried by the exchanged photon, if all particles are on-mass-shell. The information on the structure of the proton is contained completely in the q^2 dependence of the form factors.

In the present paper we report on an experiment in which the elastic electron-proton scattering has been studied. This experiment is the complement to a similar one, on which has been reported in a previous paper [3] hereafter referred to as (I). Whereas experiment (I) has been carried out at small scattering angles, the measure-

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ments have been extended to large angles in the present experiment, thus providing a considerable enlargement of the accessible kinematical region. All together the measurements have been carried out in the region of low-momentum transfer which is usually omitted in high-energy physics experiments. We present here an analysis of the data which is based on the same methods as in (I).

At low momentum transfer the contribution of the electric form factor to the cross sections dominates the magnetic contribution. As a result our measurements generally allow a better determination of $G_E(q^2)$, whereas $G_M(q^2)$ is better determined from high q^2 measurements. Another quantity which may be well determined from low q^2 measurements is the r.m.s. radius. As the r.m.s. radius is obtained from the slope of the form factor at $q^2 = 0$, it is obvious that the corresponding measurements should be carried out at the lowest possible momentum transfers. In practice the gap between the experimentally achieved momentum transfer and $q^2 = 0$ is appreciable and the form factor curve has to be extrapolated to $q^2 = 0$. Therefore, in order to obtain a reliable extrapolation, it is necessary to determine the q^2 dependence of the form factors in a rather large q^2 interval near $q^2 = 0$.

The main approach to a physical understanding of the form factors is based upon dispersion relations. In this approach the form factors are assumed to be analytic functions in the entire t plane, where t is the negative squared four momentum transfer. The form factors at space-like momentum transfer, i.e. the region which is accessible to electron-nucleon scattering, are expressed as integrals over the imaginary part of the form factor (the spectral function) in the time-like region

$$G(t) = \frac{1}{\pi} \int_{t_0}^{\infty} \frac{\operatorname{Im} G(t')}{t' - t} \, \mathrm{d}t' \,. \tag{1}$$

Physically, the spectral function is closely related to the mass spectrum of strongly interacting systems of spin and parity 1^-, baryon number and strangeness zero and of appropriate isotopic spin. The threshold t_0 is $4m_\pi^2$ for the isovector and $9m_\pi^2$ for the isoscalar intermediate states. From the vector dominance model we expect the spectral function to have peaks at the positions of the vector mesons of strangeness zero.

The very simplest assumption which can be made about the spectral function, is that it is given by a sum of single poles. In this case the integral equation of the dispersion relation (1) is solved easily and gives the Clementi-Villi type of form factors. There exists a wide variety of form factor fits which use different numbers of poles at the positions of known and hypothetical vector mesons (references may be found in the review article of Gourdin [4]). However, all these fits, which use the strengths of the poles as adjustable parameters, are successful only in a limited region of the space-like momentum transfer. Therefore in the pole fit, which has been used in the present work, additionally two of the pole masses have been varied. The resulting pole positions are in disagreement with the known vector meson masses. We believe this to be an indication that the spectral function has a considerable non-resonant contribution in addition to the prominent poles.

In order to obtain more information on the spectral function we have made an attempt to extrapolate the q^2 dependence of the experimental form factors to the time-like region of momentum transfer. As the electric form factor of the proton has been well determined by the present experiment, we have been particularly interested in the extrapolation of G_E. The spectral function is the sum of an isoscalar and an isovector part, Im G_E^S and Im G_E^V. Because of the comparatively high treshold t_0 of the isoscalar part we have concentrated on the isovector part. As a result we have obtained spectral functions Im G_E^V which show a non-resonant continuum besides a prominent ρ resonance. This feature supports the conjecture that the "unphysical" masses which have been found from our pole fits simulate the non-resonant part of the spectral function.

In sect. 2 we report on the measurements. The details of the data analysis are presented in sect. 3. In subsect. 4.1 the results of the pole fits are compared to the data of other laboratories. Subsect. 4.2 deals with the r.m.s. radius of the proton and subsect. 4.3 with the spectral function of the electric isovector form factor.

2. Measurements

The experiment was performed on the Mainz 300 MeV electron linac under conditions identical to those of (I). The target was a thin-walled cylindrical cell filled with liquid hydrogen. The target cell had a diameter of 10 mm and was mounted perpendicular to the beam axis. The kinematical region of the experiment has been extended, i.e. the scattering angle was increased from 75° to 150° and the indident electron energy was increased to 300 MeV. This has enabled us to measure the cross sections for the elastic electron-proton scattering in a q^2 range from 0.13 to 4.7 fm^{-2}.

For the measurement of the cross sections we made use of two spectrometers. The first is a conventional double-focusing 180° spectrometer which may be set to different scattering angles. The second consists of two quadrupoles and a 12° bending magnet mounted at a fixed scattering angle of 28°. The solid angle acceptance and the detection efficiency of the second spectrometer are well known (± 1%). At seven different incident electron energies an absolute cross section was first measured using the second spectrometer. The angular dependence of the cross sections was then measured setting the first spectrometer to different angles, starting with an angle of 28°. During the measurements with the first spectrometer the counting rate of the second spectrometer was also recorded and the reading of the first spectrometer was normalized to that of the second one. This was important in the present experiment because the fluctuations of the target thickness (mainly due to beam heating) amounted up to 10%, whereas the ratio of the counting rates of the two spectrometers could be measured with a typical accuracy of 1%. The determination of the absolute cross sections at 28° was not affected by beam heating effects, because these measurements have been carried out at a low beam current of 10 nA.

Table 1
The electron-proton elastic scattering cross sections obtained from the present experiment

Incident energy (MeV)	Scattering angle θ (deg)	q^2 (fm^{-2})	Cross section $d\sigma/d\Omega$ (10^{-30} cm^2/sr)	Random error (±)	$\frac{d\sigma}{d\Omega} \Big/ \frac{d\sigma}{d\Omega_D}$	Error of the ratio (±)
149.8	80.0	0.84	0.659	0.006	0.966	0.009
	90.0	0.99	0.386	0.003	0.961	0.008
	100.0	1.14	0.240	0.002	0.962	0.009
	110.0	1.27	0.1561	0.0015	0.954	0.009
	120.0	1.40	0.1089	0.0011	0.968	0.009
	130.0	1.50	0.0786	0.0007	0.968	0.008
180.1	80.0	1.19	0.437	0.006	0.963	0.013
	90.0	1.40	0.258	0.003	0.961	0.013
	95.0	1.50	0.203	0.003	0.960	0.013
	100.0	1.60	0.1621	0.0021	0.960	0.012
	105.0	1.69	0.1317	0.0018	0.962	0.013
	110.0	1.78	0.1060	0.0014	0.942	0.013
	115.0	1.86	0.0892	0.0012	0.952	0.013
	120.0	1.94	0.0749	0.0010	0.948	0.012
	130.0	2.08	0.0558	0.0006	0.953	0.010
	140.0	2.20	0.0443	0.0007	0.970	0.015
199.6	80.0	1.44	0.343	0.005	0.954	0.014
	90.0	1.69	0.205	0.003	0.960	0.014
	100.0	1.92	0.1301	0.0017	0.965	0.013
	105.0	2.03	0.1043	0.0015	0.951	0.013
	110.0	2.14	0.0856	0.0010	0.945	0.011
	115.0	2.24	0.0715	0.0009	0.943	0.011
	120.0	2.33	0.0615	0.0007	0.955	0.011
	130.0	2.49	0.0458	0.0003	0.948	0.006
	140.0	2.63	0.0359	0.0004	0.939	0.011
228.9	28.0	0.31	24.5	0.5	0.965	0.019
	40.0	0.60	5.46	0.07	0.971	0.013
	45.0	0.74	3.23	0.04	0.955	0.011
	50.0	0.88	2.04	0.02	0.957	0.011
	55.0	1.04	1.309	0.015	0.936	0.011
	60.0	1.19	0.896	0.008	0.944	0.009
	70.0	1.53	0.453	0.004	0.953	0.009
	75.0	1.69	0.333	0.003	0.955	0.008
	80.0	1.85	0.250	0.003	0.956	0.013
	90.0	2.16	0.1490	0.0021	0.957	0.013
	110.0	2.72	0.0638	0.0009	0.949	0.013
	120.0	2.96	0.0459	0.0006	0.949	0.012
	130.0	3.16	0.0351	0.0005	0.953	0.013
	140.0	3.32	0.0286	0.0004	0.968	0.012

Table 1 (cont.)

Incident energy (MeV)	Scattering angle θ (deg)	q^2 (fm^{-2})	Cross section $d\sigma/d\Omega$ (10^{-30} cm^2/sr)	Random error (±)	$\frac{d\sigma}{d\Omega} / \frac{d\sigma}{d\Omega_D}$	Error of the ratio (±)
249.6	28.0	0.36	20.40	0.40	0.967	0.019
	35.0	0.55	7.97	0.09	0.968	0.011
	40.0	0.71	4.51	0.06	0.971	0.013
	45.0	0.87	2.69	0.03	0.966	0.011
	50.0	1.04	1.670	0.023	0.955	0.013
	55.0	1.23	1.090	0.015	0.952	0.014
	60.0	1.41	0.741	0.008	0.956	0.013
	65.0	1.60	0.522	0.007	0.966	0.014
	70.0	1.79	0.369	0.004	0.954	0.011
	75.0	1.98	0.273	0.004	0.963	0.013
	80.0	2.17	0.1978	0.0030	0.929	0.014
	90.0	2.53	0.1197	0.0018	0.944	0.014
	100.0	2.86	0.0760	0.0013	0.937	0.016
	110.0	3.16	0.0521	0.0009	0.941	0.016
	120.0	3.43	0.0374	0.0006	0.932	0.016
	130.0	3.66	0.0289	0.0004	0.939	0.013
	140.0	3.84	0.0240	0.0004	0.968	0.015
275.3	35.0	0.67	6.43	0.07	0.968	0.010
	40.0	0.85	3.59	0.04	0.963	0.010
	45.0	1.05	2.11	0.02	0.949	0.011
	55.0	1.48	0.862	0.009	0.949	0.010
	80.0	2.59	0.1568	0.0016	0.937	0.009
	90.0	3.01	0.0941	0.0010	0.943	0.010
	100.0	3.40	0.0598	0.0006	0.934	0.010
	110.0	3.75	0.0413	0.0004	0.939	0.009
	120.0	4.06	0.0304	0.0003	0.948	0.009
	130.0	4.32	0.0236	0.0002	0.953	0.009
	140.0	4.53	0.01921	0.00019	0.956	0.009
	150.0	4.69	0.01633	0.00016	0.952	0.009
298.5	28.0	0.52	13.92	0.15	0.967	0.010
	30.0	0.59	10.34	0.11	0.963	0.010
	35.0	0.78	5.31	0.06	0.958	0.010
	40.0	1.00	2.96	0.03	0.956	0.010
	45.0	1.23	1.750	0.018	0.951	0.010
	50.0	1.47	1.090	0.012	0.949	0.010
	55.0	1.72	0.703	0.007	0.941	0.010
	60.0	1.97	0.472	0.005	0.939	0.010
	65.0	2.23	0.328	0.004	0.940	0.010
	70.0	2.49	0.234	0.002	0.940	0.010

$(d\sigma/d\Omega)/d\sigma/d\Omega_D$ is the ratio of the measured cross sections to the predictions of the dipole fit.

Data corrections have been applied in the same way as in (I). On account of the enlarged scattering angle the scattering amplitude correction according to formula (C.11) of Mo and Tsai [5] has been taken into account explicitly. This correction amounted to at most 0.6%. The cross sections and their random errors are given in table 1. The error in the overall normalization of the cross sections, discussed in (I), is not included in table 1.

3. Data analysis

As in (I) the q^2 dependence of the electromagnetic form factors $G_E(q^2)$ and $G_M(q^2)$ of the proton has been determined from Rosenbluth plots and by an appropriate parametrization of the form factors. Both methods refer directly to the measured cross sections.

3.1. Rosenbluth plots

The method of Rosenbluth plots is based upon the fact that the Rosenbluth formula [1] contains a linear combination of the squared form factors. In the Rosenbluth plot, cross sections which have been measured at identical momentum transfer but at different angles should be represented by a straight line. The Rosenbluth formula is given by

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}(E_0,\theta) = \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}(E_0,\theta)|_{\mathrm{NS}}\,[a(q^2)G_E^2(q^2) + b(q^2,\theta)G_M^2(q^2)]\ ,$$

$$a(q^2) = \frac{1}{1+\tau}\ , \qquad b(q^2,\theta) = \mu^2\left(\frac{\tau}{1+\tau} + 2\tau\tan^2\tfrac{1}{2}\theta\right)\ ,$$

$$\tau = \frac{q^2}{4M^2}\ , \qquad G_E(0) = G_M(0) = 1\ , \tag{2}$$

where $\mathrm{d}\sigma/\mathrm{d}\Omega(E_0,\theta)|_{\mathrm{NS}}$ is the cross section for the elastic electron scattering from a point-like and spinless particle with the mass of the proton at incident electron energy E_0 and scattering angle θ. $G_E(q^2)$ and $G_M(q^2)$ are the Sachs form factors [2] which depend only on the squared four-momentum q^2. They are normalized to 1 at $q^2 = 0$ in our notation. μ is the total magnetic moment of the proton. M is the proton mass.

From the combined cross sections given in (I) and in the present work, Rosenbluth plots have been obtained at q^2 values from 0.35 to 3.15 fm^{-2}. As the cross sections at large scattering angles (present experiment) have been measured with the same accuracy as those at small scattering angles (I), we were now able to determine the form factors better than in (I). The resulting values for G_E and G_M are listed in table 2. In addition table 2 contains the ratios G_E/G_D and G_M/G_D of the

Table 2
Electric and magnetic form factors of the proton and their ratio to the dipole fit

q^2 (fm^{-2})	(GeV/c)^2	G_E	G_E/G_D	G_M	G_M/G_D
0.35	0.0136	0.949 ± 0.008	0.986 ± 0.008		
0.55	0.021	0.930 ± 0.008	0.986 ± 0.008	0.924 ± 0.057	0.982 ± 0.060
0.86	0.033	0.901 ± 0.007	0.988 ± 0.008	0.895 ± 0.036	0.982 ± 0.040
1.03	0.040	0.874 ± 0.008	0.975 ± 0.009	0.899 ± 0.018	1.002 ± 0.020
1.18	0.046	0.859 ± 0.006	0.974 ± 0.007	0.877 ± 0.022	0.996 ± 0.025
1.40	0.055	0.840 ± 0.004	0.974 ± 0.005	0.852 ± 0.014	0.987 ± 0.015
1.70	0.066	0.810 ± 0.008	0.968 ± 0.009	0.831 ± 0.011	0.993 ± 0.012
2.16	0.084	0.772 ± 0.006	0.965 ± 0.007	0.788 ± 0.007	0.985 ± 0.010
2.55	0.099	0.746 ± 0.007	0.969 ± 0.009	0.748 ± 0.011	0.974 ± 0.015
3.15	0.123	0.693 ± 0.013	0.954 ± 0.018	0.716 ± 0.014	0.986 ± 0.020

The form factors have been determined from the cross sections quoted in this paper and in (I) using Rosenbluth plots; the errors of the form factors are due to the random errors of the cross sections (note that in our notation $G_E(0) = G_M(0) = 1$).

form factors to the corresponding values of the dipole fit (eq. (2) of (I)). The errors quoted for G_E and G_M are due to the random errors in the cross sections and have been calculated from the error matrix of the fitting routine. Taking into account the overall normalization error of the cross sections would give rise to an additional uncertainty of ±1.8% in the form factors.

Fig. 1 shows the ratio G_E/G_D *versus* q^2 determined in this work and in experiments at Erevan [6], Saskatoon [7] and Orsay [8,9]. The points of the present experiment deviate significantly from the dipole fit with a maximum deviation of about 4% at 3 fm^{-2}. This confirms the deviations from the dipole fit which have been found in (I). If one takes into account the experimental normalization errors it is seen that the measurements of the individual laboratories are not in disagreement. However, it should be noted that the measurements of the Erevan group have been done in eight q^2 intervals which individually have been normalized to a dipole formula.

Fig. 2 shows the ratio G_M/G_D determined in this work and in experiments at Orsay [8–10]. The magnetic form factor shows the same trend as the electric one. However, because of the large errors in G_M the deviation from the dipole fit is not as clear as in the case of G_E. Therefore no statements about the validity of the scaling law $G_E(q^2) = G_M(q^2)$ can be given. The uncertainty of the magnetic form factors is due to the small value of b in the Rosenbluth formula (see eq. (2)). In the present experiment even at the largest scattering angles the ratio b/a was 0.37 dropping to 0.007 at the smallest angles. A more precise determination of G_M would require a measurement of the electron-proton elastic scattering cross sections near 180°, similar to that carried out by Ganichot et al. [10]. This kind of measurement, however, cannot be made with the existing experimental setup at the Mainz 300 MeV electron linac.

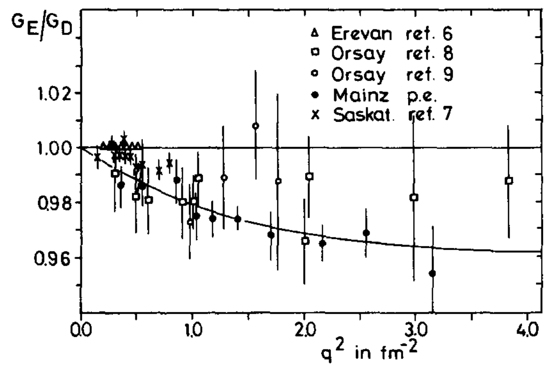

Fig. 1. The ratio G_E/G_D of the electric form factor of the proton to the dipole fit *versus* q^2. The solid circles are the points determined in the present experiment from Rosenbluth plots (cf. table 2). The error bars at the Erevan points are smaller than the symbols. The solid curve has been calculated from the best four-pole fit to the cross sections (eq. (4) with parameters of table 4).

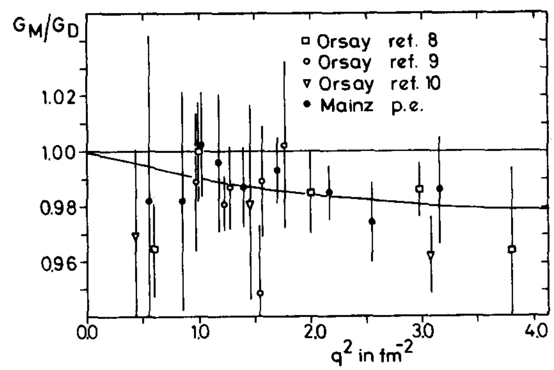

Fig. 2. The ratio G_M/G_D of the magnetic form factor of the proton to the dipole fit *versus* q^2. (Note that in our notation $G_M(0) = 1$.) The solid circles have been determined in the present experiment from Rosenbluth plots (cf. table 2). The solid curve has been calculated from the best four-pole fit to the cross sections (eq. (4) with parameters of table 4).

3.2. Form factor parametrization

Another way to extract the proton form factors $G_E(q^2)$ and $G_M(q^2)$ from the measured cross sections is to use parametrized analytical functions for the form factors in the Rosenbluth formula eq. (2). The parameters are fitted directly to the cross sections. The parametrized function should be capable of fitting all the cross sections measured at the different laboratories. Such a parametrization should be applicable over a wide q^2 range and is therefore expected to give a reliable extrapolation to $q^2 = 0$. A good extrapolation is essential both for the determination of the electric and magnetic radii of the proton and for a check of the overall normalization of the data. Generally G_E and G_M cannot be determined independently because the form factors are functions of the single variable q^2 whereas the cross sections depend strongly on the scattering angle (see eq. (2)). By fitting a large q^2 range, however, the influence of the correlation is considerably reduced.

Contrary to the method of Rosenbluth plots the method of form factor parametrization allows the q^2 behaviour of the form factors to be determined from the cross sections in a one-step process. Moreover the method of form factor parametrization exploits the whole kinematical region of a given experimental setup. (In our case the cross sections have been taken between 0.13 and 4.7 fm^{-2} whereas Rosenbluth plots could be drawn only between 0.35 and 3.15 fm^{-2}.)

In (I) a successful parametrization has been obtained which is based on the vector-dominance model. The four independent isoscalar and isovector form factors G_E^S, G_E^V, G_M^S and G_M^V of the nucleon are given in the vector-dominance model by the following expression (for details and references see ref. [4])

$$G(q^2) = \sum_V f_{VN\bar{N}}(q^2) \frac{m_V^2}{f_V} \frac{1}{\Delta_V(q^2)} . \tag{3}$$

The sum is extended over all the vector mesons V. $f_{VN\bar{N}}(q^2)$ is the vector-meson-nucleon coupling constant. m_V^2/f_V is the photon-vector-meson coupling constant and $1/\Delta_V(q^2)$ is the vector meson propagator.

As all of the quantities in eq. (3) are not known theoretically, the following simple assumptions have been made:

(i) The vector-meson-nucleon coupling constants $f_{VN\bar{N}}$ have been taken to be independent of q^2. They have been combined with the photon-vector-meson coupling constants m_V^2/f_V to give the constants $a_i m_i^2$. These constants may be different for G_E and G_M, as no scaling law for G_E and G_M is assumed.

(ii) The vector meson propagators $1/\Delta_V(q^2)$ have been replaced by simple single poles $1/(m_i^2 + q^2)$, where the finite width of the vector mesons has been neglected.

(iii) The isoscalar and the isovector parts of G_E and G_M are not normalized independently.

Making use of these assumptions we have obtained from eq. (3) the expression

$$G_{E,M}(q^2) = \sum_{i=1}^{n} \frac{a_i^{E,M}(m_i^{E,M})^2}{(m_i^{E,M})^2 + q^2} , \quad \sum_{i=1}^{n} a_i^{E,M} = 1 , \quad G_E(0) = G_M(0) = 1 , \tag{4}$$

which has been used as a parametrization of the form factors.

If we take for the quantities m_i the masses of the four vector mesons ρ, ρ', ω and ϕ and vary the strengths a_i of the poles, eq. (4) does not yield a good fit. In order to achieve a statistically acceptable fit over a wide q^2 range while restricting the number of poles to $n = 4$ (four-pole fit), we had to use two of the masses m_i as adjustable parameters. The masses of ρ and ρ' have been kept fixed. We therefore are left with a total of 10 adjustable parameters (5 for each form factor) which are fitted to the measured cross sections by minimizing the following χ^2 function:

$$\chi^2 = \sum_i \left[\frac{\sigma^i_{\text{exp}} - \sigma^i_{\text{theor}}}{\Delta\sigma^i_{\text{exp}}}\right]^2 . \tag{5}$$

The fitting routine is intricate because the cross sections do not depend linearly on the parameters. Moreover the parameters are highly correlated because the cross sections are given by a linear combination of the squared form factors, which in turn are given by four-fold sums.

The relatively large number of adjustable parameters is counterbalanced by the large number (162) of data points. Actually the fit has been carried out using in the region of low-momentum transfer the cross sections measured at Mainz and in the region of high momentum transfer the cross sections measured at Bonn [11], DESY [12] and SLAC [13] (for details see table 3). We used only data which have been obtained by means of the same radiative corrections to avoid systematic de-

Table 3
List of laboratories and q^2 ranges for the cross sections which have been fitted

Laboratory	q^2 range (fm^{-2})	Number of data points	χ^2 per data point
Mainz p.e.	0.13– 4.69	107	0.44
Bonn ref. [11]	25.0 – 50.0	20	0.60
DESY ref. [12]	17.2 – 77.1	21	0.77
SLAC ref. [13]	25.7 –642.6	14	0.64

The number of data points which have been taken into account and the value of χ^2 per data point is given for each data set (the Bonn data which have been measured at q^2 values below 25 fm^{-2} have not been fitted).

Table 4
Best fit parameters of the four-pole fit eq. (4)

a_i^{E}	$(m_i^{\text{E}})^2$ (fm^{-2})	a_i^{M}	$(m_i^{\text{M}})^2$ (fm^{-2})
0.219	3.53	0.794	8.73
1.371	15.02	0.594	15.02
−0.634	44.08	−0.393	44.08
0.044	154.20	0.005	355.40

The fit has been based on the cross section data sets listed in table 3 and has been carried out by minimizing the χ^2 function (5).

viations of the individual data sets. In (I) an additional parameter norm_L had been introduced to correct for eventual normalization differences of the individual laboratories. However, it has been found that the norm_L did not deviate significantly from unity and we decided to drop them.

The best fit parameters of the four-pole fit eq. (4) are given in table 4. The corresponding form factors are plotted as solid lines in figs. 1 and 2. It is seen that the results of the analysis using Rosenbluth plots agree with the results of the analysis using form factor parametrization. A comparison of table 4 with table 3 of (I) shows that the masses of the poles have retained their values whereas their strengths have changed slightly, mainly due to taking into account the high q^2 data of SLAC [13] which have not been included in our previous paper (I). In table 4 no parameter errors were given. Here, it was felt that a meaningful error analysis could not be performed because of the large number of parameters and their strong correlations.

4. Results and conclusions

4.1. Comparison with other laboratories

All the recently published electron-proton cross sections [8,10–18] are compared to our best fit in fig. 3, where the ratio of the measured cross sections to the corresponding fit values is plotted as a function of q^2. The fit values have been calculated from the Rosenbluth formula eq. (2) using eq. (4) with the parameters of table 4 for the form factors. The data of the individual laboratories are shown separately. Solid symbols denote data which have been included in the fitting procedure, open symbols denote unfitted data. Fig. 4 shows the same data as fig. 3 compared however to the dipole fit. It is clearly seen that the dipole fit, contrary to our best fit, fails to describe the bulk of the data. Nevertheless it should be mentioned that some of the data sets could be brought into agreement with the dipole fit by shifting the data within their normalization errors. Particularly data sets which span only a small q^2 range are easily renormalized (i.e. without bringing about an unacceptable large χ^2), because observable systematic deviations from the dipole behaviour occur only throughout a rather large q^2 range. Another interesting feature is found from the various fit calculations carried out in this work and is also seen from figs. 3 and 4. The cross sections measured at HEPL at Stanford [14] scatter strongly relative to any analytical fit. This is worthwhile to note because the results of ref. [14] are widely used to calculate reference cross sections for electron scattering experiments on nuclei. From fig. 3 it is seen that our best fit is indeed very satisfactory and is well suited to calculate reference cross sections up to momentum transfers of 650 fm^{-2}. However, it should be kept in mind that this fit is of little use in making predictions in regions where no measurements exist.

In order to obtain a statistically acceptable fit over the whole q^2 range the mass of one of the poles must in any case lie between the two-pion threshold and the

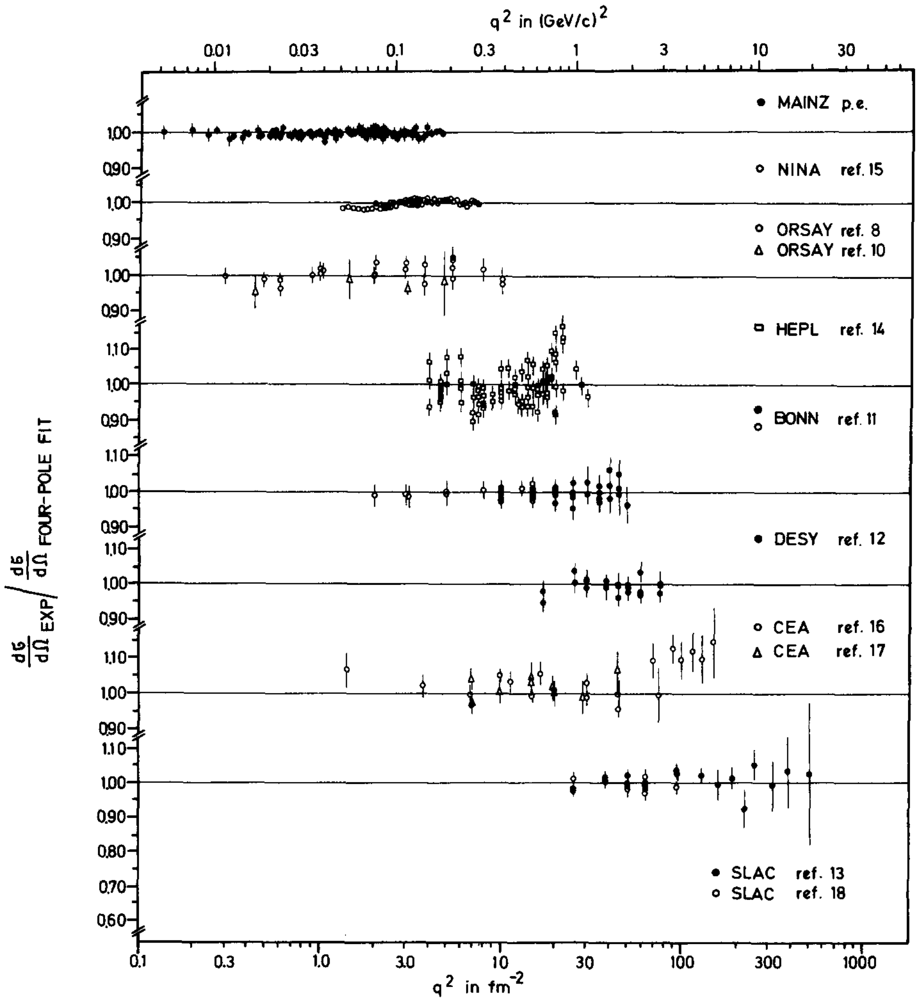

Fig. 3. The ratio of the electron-proton elastic scattering cross sections measured at various laboratories to the predictions of the best four-pole fit (eq. (4) with parameters of table 4) *versus* q^2. Solid circles denote data which have been fitted, open symbols denote unfitted data. The error bars contain no normalization errors. The points measured at NINA have been taken from fig. 1 of ref. [15]. The Bonn data which are denoted by open circles have been increased by 2% (see ref. [12]).

mass of the ρ meson. Therefore we are not surprised that for instance the fits of refs. [19,20], which use only masses greater or equal than m_ρ, show to some extent considerable deviations from the measured data. Compared to the pole fits

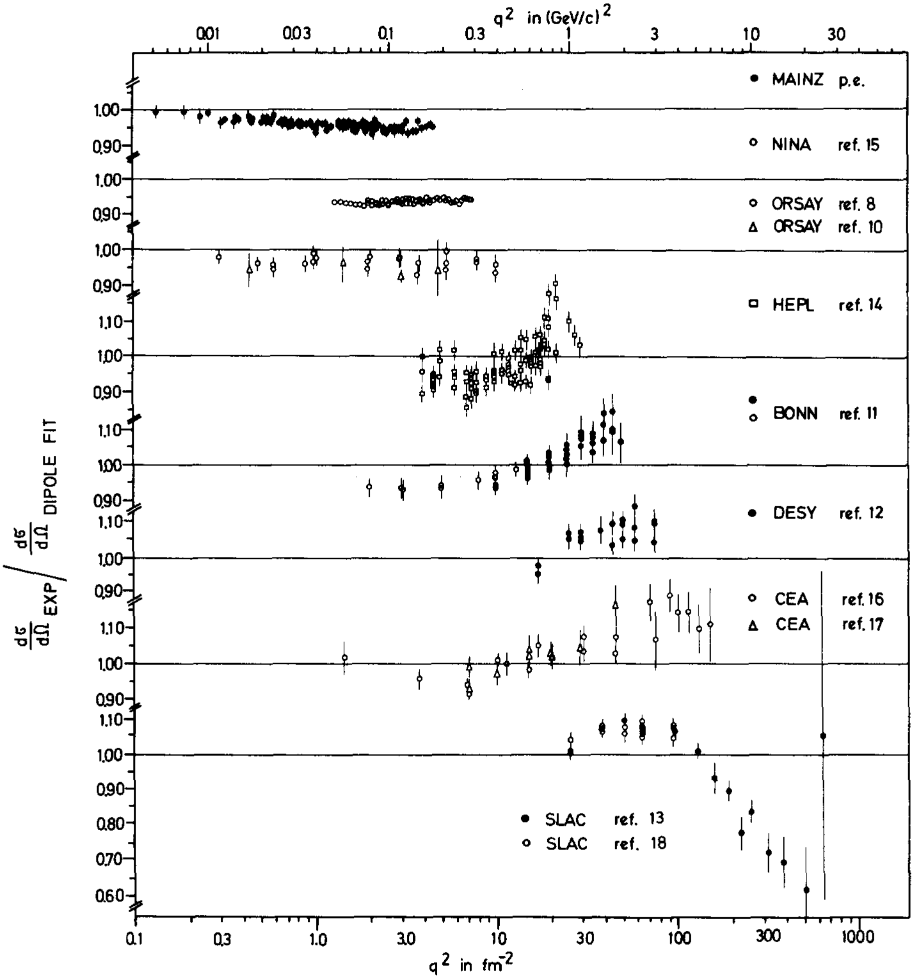

Fig. 4. The ratio of the electron-proton elastic scattering cross sections measured at various laboratories to the predictions of the dipole fit *versus* q^2. The points measured at NINA have been taken from fig. 1 of ref. [15]. The Bonn data which are denoted by open circles have been increased by 2% (see ref. [12]).

which use "realistic" masses for the poles, the low mass in our fit allows no direct physical interpretation. The need for a low mass pole originates perhaps from an enhancement of the spectral function of the proton isovector form factor due to a nonresonant two pion contribution near threshold recently reported by Höhler and Pietarinen [21].

4.2. The proton r.m.s. radius

The electric and the magnetic r.m.s. radius $\langle r_E^2 \rangle^{\frac{1}{2}}$ and $\langle r_M^2 \rangle^{\frac{1}{2}}$ of a spherical symmetric particle or nucleus are defined as the square root of the second moment of the radial distribution $\rho(r)$ of the electric charge density and of the magnetization density. In the limit of vanishing energy transfer in the scattering process (i.e. $q^2 = |\boldsymbol{q}|^2$), the proton form factors $G_E(q^2)$ and $G_M(q^2)$ are the Fourier transforms of the corresponding radial distributions $\rho_E(r)$ and $\rho_M(r)$. For higher values of the momentum transfer the Fourier transform is no longer defined and the form factors may not be interpreted in terms of radial distributions. In the low q^2 case the Fourier integral may be developed in terms of $|\boldsymbol{q}|^2$

$$G_{E,M}(q^2) = 1 - \tfrac{1}{6}\langle r_{E,M}^2 \rangle |\boldsymbol{q}|^2 + \tfrac{1}{120}\langle r_{E,M}^4 \rangle |\boldsymbol{q}|^4 - + \dots \,, \tag{6}$$

where one of the expansion coefficients is essentially the r.m.s. radius. By fitting the expression (6) to the form factors the r.m.s. radius is obtained model independently.

The form factors which have been used to determine the r.m.s. radius have been extracted directly from the low q^2 cross sections ($q^2 < 0.9$ fm^{-2}) of (I) and of the present experiment under the assumption $G_E = G_M$. For the extraction of G_E only those cross sections have been taken into account to which G_M contributed not more than 10% (i.e. $b(q^2, \theta)/a(q^2) \leqslant 0.1$ in the Rosenbluth formula eq. (2)), so that any deviation from $G_E = G_M$ would be suppressed by at least a factor of 10. Since there is no evidence for a significant violation of the $G_E = G_M$ law at very low momentum transfer [22,23], the direct extraction of G_E from the cross sections is believed to give reliable results in this region. The G_E values which have been obtained from this procedure are plotted in fig. 5 as a function of q^2.

For $q^2 < 0.9$ fm^{-2} the contributions of the higher terms in the expansion (6) are negligable and the series can be truncated to give $G_E(q^2) = \delta + \beta q^2$. From fitting this expression to the form factors of fig. 5, the solid line of fig. 5 has been obtained. The best fit parameters were $\delta = 0.994 \pm 0.002$ and $\beta = -0.118 \pm 0.004$ fm^2. The reduced χ^2 was 0.5. The result of the fit did not depend significantly on the fitted q^2 range. This was checked by fitting additionally the G_E values of table 2 up to 1.2 fm^{-2}. The addition of a q^4 term to the fit formula did not improve the fit, moreoever the error of the additional parameter turned out to be larger than its value. The best fit value of the parameter δ is well within the normalization error of the G_E values. The best fit value of the parameter β gives a proton r.m.s. radius of $\langle r_E^2 \rangle^{\frac{1}{2}} = 0.84 \pm 0.02$ fm. This value is higher than the dipole value of 0.81 fm, but within the error limits it is compatible with the result (0.81 ± 0.04 fm) of a similar experiment carried out at Saskatoon [7]. The radius values reported in (I) resulted from the model-dependent extrapolation of the form factors to $q^2 = 0$, where the models demand $G_{E,M}(0) = 1$. If we include this demand in the above described fit

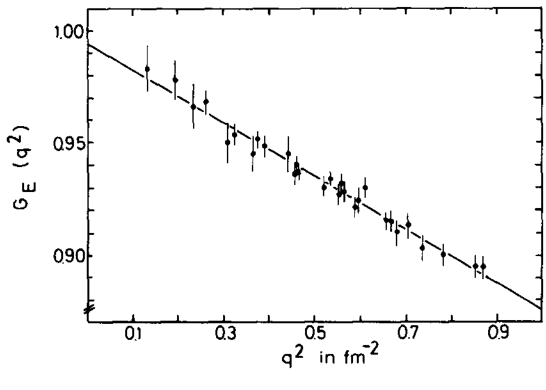

Fig. 5. Form factor values which have been extracted directly from the cross sections of (I) and of the present experiment under the assumption $G_E = G_M$. The contribution of G_M to the cross sections was less than 10%. The solid line is the best fit using the formula $G_E(q^2) = \delta + \beta q^2$.

(i.e. $\delta = 1.0$), we obtain a proton r.m.s. radius of 0.88 fm, a value which coincides with the extrapolation results.

4.3. Extrapolation of the proton form factors to the time-like region of momentum transfer

The extrapolation of the experimental form factors to the time-like region of momentum transfer has been carried out by means of the conformal transformation technique which has been used for instance by Levinger and Peierls [24]. In this transformation the cut t plane is mapped conformally onto the interior of the unit circle

$$z(t) = \frac{\alpha - \sqrt{t_0 - t}}{\alpha + \sqrt{t_0 - t}}, \tag{9}$$

where α is a constant with typical values between 0.5 and 2.0 GeV/c. t_0 is the value of the threshold. Following a suggestion of Höhler [25] we used as an ansatz for the isovector part of the nucleon form factor

$$G^V(z) = \frac{(1+z)^4}{(z-z_\rho)(z-z_\rho^*)} \left\{ \frac{[z(0)-z_\rho]\,[z(0)-z_\rho^*]}{[1+z(0)]^4} + [z - z(0)] \sum_{n=0}^{N} c_n z^n \right\},$$

$$z_\rho = \frac{\alpha + [4m_\pi^2 - (m_\rho - \frac{1}{2}i\Gamma_\rho)^2]^{\frac{1}{2}}}{\alpha - [4m_\pi^2 - (m_\rho - \frac{1}{2}i\Gamma_\rho)^2]^{\frac{1}{2}}}. \tag{10}$$

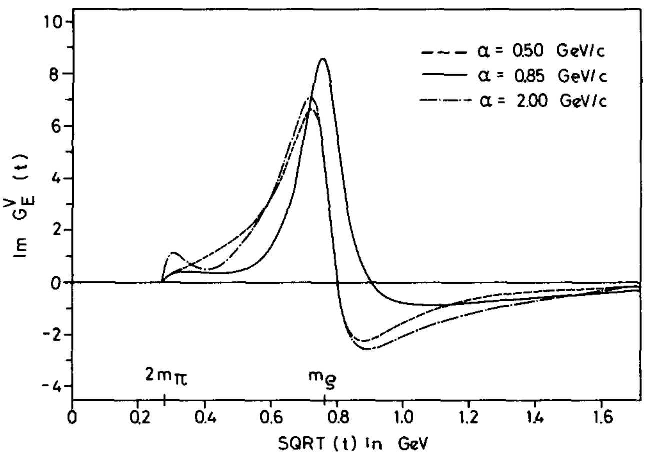

Fig. 6. Spectral functions of the electric isovector form factor of the nucleon. The curves have been obtained from a three-parameter ansatz of the photon-nucleon coupling which incorporates a realistic ρ pole using the technique of conformal mapping. The curves are best fits to the proton form factors measured in the space-like region of momentum transfer. They differ because of the choice of the constant α in the transformation eq. (6).

As there is much experimental evidence that the isovector spectral function is dominated by the ρ resonance, eq. (10) incorporates a realistic ρ pole at $m_\rho = 765$ MeV with a width of $\Gamma_\rho = 135$ MeV. The remaining isovector spectral function is given by a polynomial in z with adjustable parameters c_n. The spectral function (10) is analytical inside the unit circle and fulfills the constraints $G^V(t=0) = 1$ and $G^V(|t| \to \infty) \sim t^{-2}$. Moreover it behaves like a p-wave at the two-pion threshold as is demanded by angular momentum conservation. For the isoscalar part of the nucleon form factor we used a pole ansatz, because the isoscalar threshold is comparatively high and because the isoscalar resonances are sufficiently narrow to approximate them by poles. The pole ansatz had two poles at the positions of the ω and the ϕ resonance

$$G^S(t) = \frac{a_\omega m_\omega^2}{m_\omega^2 - t} + \frac{a_\phi m_\phi^2}{m_\phi^2 - t}, \qquad t \leqslant 0 . \tag{11}$$

The strengths a_ω and a_ϕ of the poles have been adjusted. The sum of eqs. (10) and (11) has been used to fit the electric form factors obtained in this experiment and from other measurements [11]. In order to obtain a good fit it was sufficient to use a polynomial of second order ($N = 2$) in the spectral function (10).

The form of the resulting electric isovector spectral function Im G_E^V depends strongly on the choice of the parameter α of eq. (9) which governs the angular position of the ρ pole in the z plane. Fig. 6 shows the best fit spectral functions for three different values of α. Although the curves differ appreciably, they all describe the measured proton form factors satisfactorily. The curves in fig. 6 are similar to those found by Levinger [26] for the magnetic isovector spectral function of the nucleon. This is not surprising because there is no experimental evidence that the scaling relation $G_E(q^2) = G_M(q^2)$ is significantly violated. The spectral functions found here and by Levinger [26] show a considerable non-resonant background between the two-pion threshold and the ρ pole, which is also indicated by the low mass of one of the poles in the four-pole fit (see table 4). The actual form of this background, however, is uncertain. This lack of knowledge is, in our opinion, mainly due to the fact that G_E^V is only one out of four nucleon form factors and cannot be uniquely determined from electron-proton scattering data alone. It is for this reason that we shall measure the electron-deuteron scattering in the near future to obtain at least indirectly further information on the neutron form factors and hence on G_E^V.

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* From comparison with the doctoral dissertation of Harms [12a] it is found that table 3 of ref. [12] contains three misprints. In row 13 the scattering angle should read 14.0° and the cross section should read 8.442 nb/sr. In row 16 the scattering angle should read 15.0°.

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